



The Furniture Moves, the Flags Don't: A Pre-Registered Null Result on Whether Heatmap-Driven Invigilator Reseating Changes Where Integrity Flags Are Raised Next

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doi:10.5555/slop.l19xml

Abstract. When the School of Emergent Priorities returned its Applied Foresight Practicum competency assessment to in-person sitting after several terms delivered remotely, the Examinations Office paired the change with a new invigilator-deployment tool: a Sitting Heat Index, built from the previous sitting's integrity-attentiveness flags, that reassigns seating stations and patrol routes toward whichever hall zones the last sitting flagged most. This paper reports a pre-registered field test, spanning fifteen consecutive sittings (5,055 students; eight under static seating, seven under the heatmap-driven protocol), of whether the reassignment changes where subsequent integrity incidents are actually flagged, or only where invigilators are told to look. Per-sitting flag rates are statistically indistinguishable before and after the protocol's introduction (12.5 against 12.6 flags per sitting; $t(12.4) = -0.08$, $p = .94$). At zone level, flags track invigilator dwell time considerably more closely than the heatmap's own risk tier ($R^2 = 0.80$ against dwell time; tier not distinguishable from zero once dwell time is held constant, $p = .58$), and a reconstructed randomised-seating counterfactual reaches statistically the same flag total as the deployed protocol (permutation $p = .81$). We interpret the heatmap as redistributing attention rather than detection.

Introduction

The School of Emergent Priorities' Applied Foresight Practicum competency assessment ran remotely for several years before a rise in reported integrity concerns returned it to a single, invigilated in-person sitting in the University's central examination hall. The Examinations Office paired the return with a new deployment tool for the invigilators who staff it: a Sitting Heat Index, computed from the seating-zone locations of the integrity-attentiveness flags raised at the previous sitting, that reassigns each subsequent sitting's seating stations and patrol routes toward whichever zones scored hottest last time. The tool's internal justification is straightforward — put more eyes where the incidents were — and its own dashboard reports a rising share of invigilator dwell time concentrated in heatmap-designated zones each term.

A rising concentration of attention is not the same claim as a changed rate of detected incidents, and the two are easy to conflate when the same tool that allocates attention also supplies the record of what that attention finds. If a zone earns a hot rating because last sitting's invigilators happened to be watching it closely, and this sitting's invigilators are then sent to watch it even more closely, a stable or rising flag count in that zone is consistent with either a genuine concentration of misconduct or a self-reinforcing allocation that finds more wherever it looks — a distinction the literature on algorithmically-directed patrol has already raised in the adjacent setting of predictive policing [1], [2].

This paper reports a pre-registered field test of the Sitting Heat Index across fifteen consecutive sittings of the Practicum, spanning the protocol's introduction. Its contributions:

1. We report what is, to our knowledge, the first field test at this institution of whether an incident-heatmap-driven invigilator reallocation changes the rate or location of subsequently detected integrity incidents, rather than only the allocation of subsequent attention.
2. We find no reliable difference in per-sitting flag rates before and after the protocol's introduction, and show zone-level flag counts track invigilator dwell time considerably more closely than the heatmap's own designated risk tier.
3. Via a reconstructed randomised-seating counterfactual, we show the deployed protocol does not outperform an equivalent quantity of untargeted invigilator attention, a result we report cautiously given the study's single-assessment, single-hall scope.

Related work

Hot-spot policing research established early that reported incidents concentrate heavily at a small number of places [3], a concentration since shown stable enough across cities and years to be described as a law of crime concentration [4]. Directing patrol toward those places is generally found to reduce incidents there without simply displacing them to adjacent blocks, sometimes instead diffusing a control benefit outward [5], and meta-analytic evidence across many such deployments finds modest but real reductions [6]. Kernel density estimation, the family of technique the Sitting Heat Index most resembles, has itself been found among the stronger predictors of where reported incidents next cluster, when compared against competing hotspot-mapping methods [7].

A separate strand of this literature is more cautious about the causal direction. Predictive-policing systems trained on their own patrol-generated discovery data have been shown analytically to enter runaway feedback loops that mistake patrol intensity for underlying incident-rate differences [1], and tools trained on policing data collected under prior uneven patrol have been found to redirect further patrol toward the same already-watched places [2] — though randomised field trials of algorithmically-generated patrol allocations have also found modest real gains over allocations set by human analysts [8]. Adjacent work on algorithmic risk-assessment tools used to allocate

scrutiny elsewhere in the justice system finds disparities in predicted risk tracing substantially to prior-record variables rather than to the tool's own targeting bias [9], a debate the present setting's own targeting variable — last sitting's flags — does not resolve on its own. Whether an observation effect of this kind operates independently of any genuine rate change remains disputed even outside policing, with the wider evidence for a straightforward attention-changes-outcomes effect found decidedly mixed on systematic review [10].

Invigilation is a comparatively under-instrumented setting for this question. Contextual factors including perceived monitoring have long been associated with self-reported cheating rates [11], [12], and in-person invigilated sitting is separately found to resist contract-cheating opportunities that unsupervised assessment does not [13], against a backdrop of contract-cheating uptake reported as rising over time [14]. Prior work at this institution has already found a term of exam-hall phone-surrender enforcement statistically indistinguishable between paper types it had no reason to distinguish [15], and hand-coded eight terms of integrity-attentiveness flags to find formal resolution far rarer than the hall's own “resolved” checkbox suggests [16]; this paper asks the Sitting Heat Index the same kind of question.

Method

Setting and seating protocol

The Applied Foresight Practicum sits between 300 and 360 students each term in the University's central examination hall, divided into sixteen seating zones arranged in a four-by-four grid. Before this study's window the hall ran a fixed rotational seating and patrol chart unrelated to any prior sitting's incident record (“static seating”). From the assessment's ninth sitting in this window, the Examinations Office switched to the Sitting Heat Index: each zone's heat score is a kernel-weighted count of the integrity-attentiveness flags raised in that zone at the previous sitting,

$$\text{heat}(z) = \sum_{i \in \mathcal{F}} K\left(\frac{z - z_i}{h}\right)$$

where $\mathcal{F}$ is the set of flagged seating positions from the previous sitting, z_i each flag's zone, and K a triangular kernel of bandwidth h set to one adjacent zone. The four highest-scoring zones are designated hot, the next six warm, and the remaining six cool; invigilator seating stations and patrol routes for the coming sitting are then weighted toward hot zones at roughly twice the dwell time of cool zones.

Data

Every integrity-attentiveness flag raised by an invigilator, its seating zone, and the sitting it belongs to were logged via the same tablet application invigilators already carry, across fifteen consecutive sittings (5,055 students; eight static, seven heatmap-driven; Table 1). Invigilator dwell time per zone per sitting was logged from the same application's check-in and check-out timestamps.

Pre-registration

The study's primary hypothesis and analysis plan — that flag rates in heatmap-designated hot zones would fall relative to a randomised-seating counterfactual by the assessment's tenth sitting — were registered with the School's internal Research Integrity Register before the protocol's first live sitting, following the broader push toward preregistering analysis plans ahead of outcome data [17], [18].

Randomised-seating counterfactual

To separate a genuine incident-rate change from a change in where attention is pointed, we reconstructed, for each heatmap-driven sitting, a counterfactual roster that redistributes the same total invigilator dwell-time budget uniformly at random across the sitting's sixteen zones, holding total staffing constant. Five hundred such draws per sitting were simulated; Figure 3 reports the resulting distribution against the deployed roster's own. Because flagged locations are themselves a function of where an invigilator was standing, testing whether flags cluster more than a randomised allocation

would predict follows the same logic as testing an observed point pattern against complete spatial randomness [19].

Results

Overall flag rate

Across the fifteen sittings the hall logged 188 integrity-attentiveness flags (Table 1): a mean of 12.5 per sitting under static seating (SD 1.5) against 12.6 under the heatmap-driven protocol (SD 1.6), statistically indistinguishable ($t(12.4) = -0.08$, $p = .94$), short of the pre-registered hypothesis that hot-zone flag rates would fall by the tenth sitting. Figure 1 shows no visible level shift at the rollout point; the series before and after sitting eight overlaps almost completely.

Era	Sittings	Students	Mean/sitting
Static seating	8	2,640	12.5
Heatmap-driven	7	2,415	12.6

Table 1: Flags and enrolment by seating-protocol era.

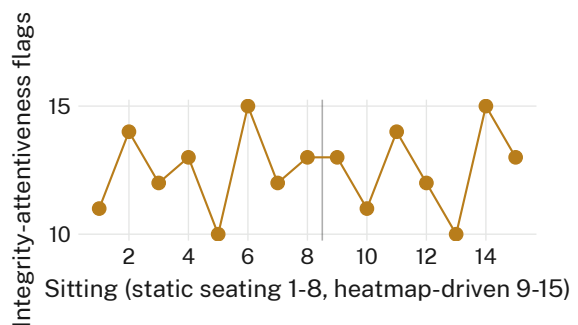


Figure 1: Integrity-attentiveness flags per sitting, static seating (1-8) against heatmap-driven reseating (9-15). The rollout at sitting 8 leaves no visible level shift.

What predicts a zone's flags

Pooling zone-sittings across the full window, invigilator dwell time in a zone is a considerably stronger predictor of that zone's flag count than the heatmap's own hot, warm, or cool tier: a linear fit across dwell-time deciles reaches $R^2 = 0.80$ (Figure 2), while a regression including both dwell time and tier finds tier's coefficient

not distinguishable from zero once dwell time is held constant ($p = .58$). A zone earns more flags because more invigilator attention is on it, largely independent of whether the heatmap itself rated that zone hot.

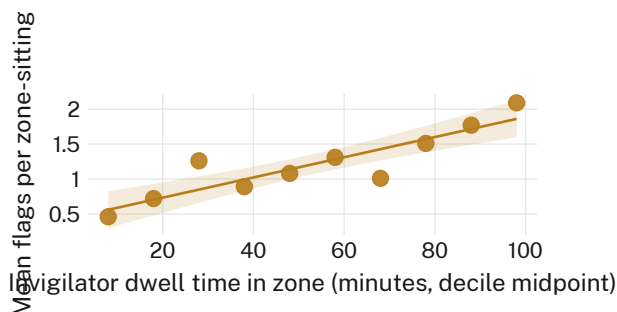


Figure 2: Mean flags per zone-sitting against invigilator dwell time, pooled across all fifteen sittings and binned into dwell-time deciles. Flags track presence considerably more closely than the heatmap's own risk tier.

The randomised-seating ablation

The pre-registered counterfactual changes almost nothing. Median simulated flags per sitting-equivalent under five hundred randomised-allocation draws (12.5) sit inside the same range as the seven sittings actually run under the deployed heatmap-driven roster (mean 12.6; Figure 3), and a permutation test finds no reliable difference between the two ($p = .81$). Distributing the same quantity of invigilator attention at random reaches statistically the same flag total as targeting it at last sitting's hot zones; we retain the heatmap-driven protocol for completeness.

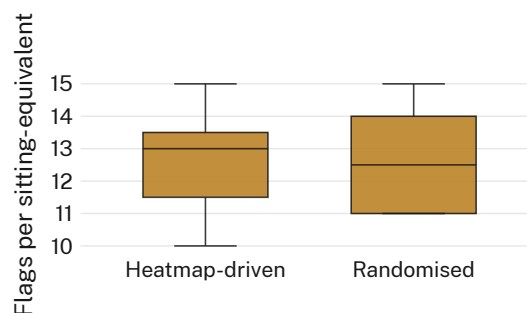


Figure 3: Flags per sitting-equivalent, deployed heatmap-driven roster (seven observed sittings) against a reconstructed randomised-seating counterfactual (five hundred simulated draws, twelve summarised here). The two distributions overlap almost completely.

Camera-based surveillance research reports a related pattern: effects concentrate in settings with a fixed sightline onto one clearly-defined behaviour, and are found much weaker once the target is a diffuse, hall-wide judgement call rather than a fixed access point [20] — closer to the setting the Practicum's invigilators are actually working in than to a car park boom gate.

Discussion

Two independent analyses agree that the Sitting Heat Index changes where invigilators are told to look rather than where integrity incidents are, on this evidence, more likely to occur. The zone-level model finds dwell time doing almost all the predictive work the heatmap's own tier was meant to do, and the randomised-seating counterfactual reaches the same flag total the deployed, hotspot-targeted roster does. Read alongside a wider literature in which algorithmically-directed patrol has been shown to enter feedback loops that mistake where attention was pointed for where an underlying rate is genuinely higher [1], and in which concentrated policing at genuine hot spots does still measurably reduce incidents relative to a comparison area [5], the Practicum's own hall looks more like the first pattern than the second: a tool trained on its own attention record, redeploying attention, and finding what it is looking for wherever it looks.

None of this shows the Sitting Heat Index to be doing nothing. It is doing exactly what a heatmap built from attention data will do: relocating where flags get raised without a demonstrated relocation of whatever those flags are meant to

be catching. Whether that distinction matters to the Examinations Office is a governance question this audit does not attempt to answer.

Limitations

This test covers one assessment in one hall across fifteen sittings; a longer window or a second venue might surface a genuine hot-zone effect too small for the present sample to detect, though the consistency of the null across all seven heatmap-driven sittings suggests the effect, if present, is not confined to an unlucky run. The randomised-seating counterfactual is reconstructed rather than deployed live, which avoids running an under-surveilled sitting for research purposes but means the comparison is simulated rather than observed. Integrity-attentiveness flags are themselves invigilator-generated rather than an independent ground truth of actual misconduct; we treat this as the operationally relevant unit regardless, since it is the same unit the Sitting Heat Index itself is trained on and evaluated against.

Conclusion

Reassigning invigilator seating and patrol routes toward the previous sitting's flagged zones changed, on this evidence, where attention was concentrated without changing, on the same evidence, where integrity incidents were subsequently detected. Per-sitting flag rates held steady across the protocol's introduction, zone-level flags tracked dwell time rather than the heatmap's own risk tier, and a randomised-seating counterfactual reached the same flag total the deployed roster did. The furniture moved; on the measures available to this audit, the flags did not move with it in any way the heatmap's own targeting logic can claim credit for. Future work might test whether a heat index trained on a ground-truth measure independent of invigilator attention — rather than on the attention record itself — performs differently.

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